
FAKE NEWS DETECTION

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ABSTRACT

Fake news, which is described by deliberately presenting false or misleading information, has become a big challenge in the digital world. This paper basically throws light on the detection and evolution of fake news, while having a focus on machine learning methodologies, natural language processing (NLP) techniques, and the latest large language model (LLM)-based solutions. Here, an explanation of traditional machine learning models like logistic regression, Naive Bayes, support vector machines, decision trees, random forests, and neural networks is provided, followed by the unsupervised learning models such as clustering, anomaly detection, and topic modeling. In addition to this, the paper also describes NLP methods including tokenization, stemming, lemmatization, word vectorization, TF-IDF, Bag of Words, and word embeddings. when it comes to fake news detection, cutting-edge LLM-based methods such as DAFND and SheepDog frameworks are compared with the traditional fake news detection methods, which is followed by the description of LLM styled attacks which make fake news look more authentic and reliable. At last, the importance of education and policy in combating fake news, along with the challenges of deepfake detection, influencer account hacking, and real-time detection, are discussed. The study basically highlights the critical need for diverse approaches, required to combat the complex issue of fake news, while laying stress on the significance of societal awareness and strong government frameworks.

Keywords :

Fake News Detection(FND), Supervised Learning Models, Unsupervised Learning Models, Natural Language Processing (NLP), TF-IDF (Term Frequency-Inverse Document Frequency), Bag of Words (BoW), Large Learning models(LLM), etc.,

1 Introduction

Fake news is basically fabricated information which appears to mimic news but it fails to provide evidence and often aims to influence public opinion. Most of the time fake news is being spread in order to create unrest in civilised society, in order to create a regime change or to financially harm a Business Firm. One way or the other, it definitely harms the principles of free speech. In recent times, Fake news has been a one of the biggest problems, and there is utmost need to find better ways to detect it and stop it, before it reaches to larger section of population. This paper will look towards a wide range of studies and methodologies which have been formulated in order to detect FND (Fake news Detection). It tells about the tools and techniques which have been previously used and about the ones which are currently developed as well. This paper throws light on what is currently happening in this area, what are the challenges and what could be potential and emerging solutions for fake news detection. By understanding this problem statement and its societal implications, a better understanding of this issue can be formulated and better ways can be derived which contribute to solve this problem effectively.

2 Literature Review on Fake News Detection

2.1 Definition and Scope of Fake News

Fake news can be defined as information or news, which is **deceptive** and false, leading to create an emotion of **mistrust** among the readers. It can't be compared with errors or opinions as it is **intentionally crafted to manipulate the public opinion**. The main characteristics of fake news include lack of **credibility** as the origin of the news remain unknown, **emotional appeal** as may exploits emotions of **masses** for **engagement** and sharing, **misleading headlines** which is designed in such a way to attract more clicks [1]. It may contain **manipulated** content like images, videos and text, which spread through social media networks at a **massive pace**.

Distribution of fake news may create **unrest** in the society as it leads to **erosion** of trust between people and institutions. Fake news spread at a massive scale **significantly** changes opinion, which **incites** further divisions in the society, further leading to conflicts and as a whole creating a polarised society. When a society is **polarised**, then people from different political mindset, prefer to vote a candidate who **aligns with that specific ideology**. Hence, it influences elections and public policy in a **drastic** manner. The following flowchart also helps to explain what fake news may potentially results in.

The **scope** of fake news is vast and comprises of a wide range of **tactics**: The following are the different types of fake news, which are evident in public domain:-

- **Deceptive News**:- This basically consists of news, which is intentionally manipulated to mislead the public [2].
- **False News**:- The completely fabricated information presented as truth, generally created to manipulate public opinion [2].
- **Satire News**:- The news which is satirical and is aimed to criticize without having any intent to harm or create discomfort is Satire News [2].
- **Disinformation**:- It is basically the spread of false or wrong information, with the intention to deceive. it is generally used to spread propaganda or to create confusion or distrust among the people [2].
- **Misinformation**:- This basically refers to false or inaccurate information, which is basically spread without the intention to deceive or harm. Most of the time, it is considered to be an outcome of misunderstanding or simply sharing inaccurate information without verifying its authenticity [2].
- **Rumour**:- The type of information which is unverified and is generally spread informally, often by word of mouth, without having any authenticate source is known as Rumour [2].

2.2 Historical Overview of Fake News and Misinformation

The term "Fake news" has become so common in **media circles**, which definitely tells about the **pervasive issue** of misinformation in the **digital age**. However, information's **distortion** is not just a recent **phenomena**, it's historical with its root extending far beyond the creation of internet [3]. From the strategic **propaganda** of ancient Rome, where Octavian **peddled false** narratives to harm his political rival Mark Antony, to the **sensationalist** and attention seeking journalism of the 19th century, the **manipulation** of public opinion through wrong or misleading **narratives** has been a consistent strategy employed by individuals and **entities** seeking to **advance their interests** [3]. The 19th century saw the rise of printing press and other forms of print media, which definitely helped to **disseminate** both **fabricated** and **factual** information. A news related to infamous "**infamous Great Moon Hoax**" which was published in New York Sun, captured the attention of readers with the stories of **lunar creatures** and demonstrated in **public domain** that how easily **sensationalised** stories could be believed and spread at a massive scale [3]. It's similar to a story which was peddled across media that tomatoes could protect from getting a serious disease, but it was completely **false narrative**, which was designed to increase the sales of tomatoes [3].

2.3 Importance of Fake News Detection

The **exponential growth** of internet and social media has **transformed** the way a fake news is spread. These days it is spread at such a **unprecedented** scale, that it becomes really difficult to **inhibit** and stop it's **propagation** through different media networks. Now ,this has created an **ecosystem**, where **combating** fake news has become a really challenging task [3]. The 2016 US Presidential election is one of the best examples of it, where fake news and **fabricated** stories were **peddled** across different media platforms, which ultimately resulted in influencing public opinion and **undermining** the trust in **traditional media platforms** [3].

These days, the **economic incentives**, which are associated with the online advertisement have contributed significantly to the rapid growth of **clickbait** and sensational content, which is often at the expense of **authenticity** and accuracy [3].

Moreover, the **evolving technologies** continues to pose huge challenge, when it comes to **curbe** the flow of **misinformation**. From the emergence of **deepfakes**, that it is highly difficult to stop the creation of deceptive content [3]. As there will be more innovation in future, It will definitely become more difficult to distinguish between genuine and fabricated content and in order to make a clear distinction, innovative approaches to detection and verification will be required [3]. Thus, there is an **utmost** need to develop technologies, that can **counter** fake news in **quick succession**.

3 Methodologies for Detecting Fake News

3.1 Machine Learning Approaches

In order to provide a detailed explanations about the machine learning algorithm's performance in fake news detection, all the algorithms can be categorised into supervised and unsupervised learning models [4]. Let's try to review the methodologies and their effectiveness, based on the literature review.

Supervised Learning Models

When the labeled data is required to train the algorithm, then that model is classified as supervised learning models. based on input-output pairs models learn the pattern and make predictions on new data [4].

a. Logistic Regression

Logistic Regression is commonly used machine learning algorithm, which is generally used in Binary Classification problems, including fake news detection as well. It predicts the probability for a model, whether it belongs to a specific class or not [4]. Generally, when the data is linearly separable, then the performance of logistic regression is good, but it struggles to make clear classification, when the data is in complex form [4].

b. Naive Bayes

The probabilistic models which are based on Baye's Theorem and in which all features are considered independent of each other [4]. This method is highly effective for text classification tasks such as the sentiment analysis and spam detection, due to the effectiveness. The performance of this algorithm is good even when the dataset is small and it provides clear baseline results, when it comes to fake news detection [4].

c. Support Vector Machines (SVM)

SVMs are powerful classifiers which are highly effective when the number of features exceed the number of the samples [4]. They work very well in higher dimensional spaces. When it comes to fake news detection, both linear and non linear data can be handled by the svm with the help of kernel functions [4]. SVMs perform very well on textual data, and their performed is increased if they are combined with feature extraction techniques like TF-IDF [4].

d. Decision Trees and Random Forests

The non paramedic models that split the data based on feature values in order to make prediction are called decision trees [4]. Random forest is an ensemble method which uses multiple decision trees and provide more accurate predictions by averaging the individual trees output's [4]. They can capture complex interactions between features and are useful in handling large datasets with variable information [4].

e. Neural Networks and Deep Learning Models

Neural networks like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), have been highly effective when it comes to fake news detection [4]. CNNs captures spatial hierarchies in data, which is highly beneficial for text data analysis whereas RNNs are good at handling sequential data, which makes them suitable for understanding the context sequences in the text [4]. Deep learning models, like BERT (Bidirectional Encoder Representations from Transformers), have enhanced the accuracy by understanding word embeddings in contextual form. Due to this reason, they are considered state-of-the-art in NLP tasks, even in fakes news detection as well [4].

Unsupervised Learning Models

Unsupervised form of machine learning is the one which don't require labelled data. In this, basically patterns and structures in the input data are identified, without any information related to the labels [4].

3.1.1 Clustering

In this form of unsupervised learning model, basically similar data points are clustered together. K-Means clustering and Hierarchical Clustering are quite famous unsupervised ml algorithms [4]. When it comes to fake news detection, clustering has proved to be effective in recognising patterns in news articles which may indicate false information, without the help of labels [4]. Even though, it has given good results, but still clustering alone is not sufficient for accurate classification and it requires post-processing or integration with other supervised methods.

3.1.2 Anomaly Detection

This technique is used to identify unusual patterns or outliers in the data. In fake news news detection, such methods are quite useful in detecting news articles which significantly, deviate from standard reporting, which ultimately suggests that something is wrong with this news, thus suggesting potential misinformation [4]. these can be combined with other methods to detect fake news [4].

3.1.3 Topic Modeling

Topic modeling techniques, such as Latent Dirichlet Allocation (LDA), can definitely identify hidden topics in a set of documents [4]. When it comes to fake news detection, topic modelling is highly useful in understanding the themes in news articles and can accurately identify the topics which are suspicious and are frequently associated with fake news [4].

3.2 Natural Language Processing (NLP) Techniques

Natural Language Processing (NLP) is a field of artificial intelligence which basically focusses on the interaction between human languages and computers [5]. In Natural language processing, basically large amount of natural language data is analysed and processed, which further enables the machines to understand and respond to human languages. When it comes to the detection of fake news, NLP is utilised in order to identify misleading or false information via textual analyzation [5].

Tokenization, Stemming, and Lemmatization

- **Tokenization:** This basically includes the breaking down of text or string into smaller units which are known as tokens (can be words, phrases, or symbols) [5]. For example, the sentence "Fake news spreads quickly" can be tokenized into ["Fake", "news", "spreads", "quickly"] [5]. In order to numerically analyse a text, tokenization needs to be performed [5].
- **Stemming:** After the removal of suffixes, the words are reduced to their root or base form [5]. For example, "singing," "swinger," and "sang" all reduce to "sing." This is a simple process and may result in production of non words [5].
- **Lemmatization:** This process is used when we need to reduce the words to their base or dictionary form, which also known as lemma [5]. For example, "better" becomes "good," and "am," "is," and "are" become "be." Lemmatization helps in maintaining base forms in the meaningful manner [5].

Word Vectorization

Textual data is transformed into numerical vectors, in order to to allow machine learning models to process and analyse it, through Word Vectorization [5]. This transformation is super important as models can only operate on numerical data, not on raw text [5]. Following are the main vectorisation techniques:-

TF-IDF (Term Frequency-Inverse Document Frequency)

This method measures the importance of a word in a document, which is further related to a collection of documents (known as corpus) [5]. TF-IDF is calculated by multiplying term frequency by the inverse document frequency [5]. Term frequency stands for how often a word appears in a document and inverse document frequency is a measure of how common or rare a word is across all documents. TF-IDF is quite useful in highlighting terms which may be more informative, when it comes to differentiate fake and real news [5].

3.2.1 Bag of Words (BoW)

In this technique, text is classified as collection of words without taking structure or order into consideration [5]. Representation of words is done, via vector of word counts. This technique doesn't capture word order or relationships, thus it may lose its important context and meaning [5].

Method	Advantages	Disadvantages
TF-IDF	The TF-IDF model includes information on both the more significant and less important words.	Slow for large vocabularies. Does not capture position in text, semantics, co-occurrences in different documents, etc.
Bag-of-words	The ease of implementation.	It ignores the ordering of the words in a given document. Ignores the semantic relations among words.
Word2Vec	Maintains the semantic meaning of various words in a text. The context information is preserved. The size of the embedding vector is very small.	Inability to deal with unfamiliar words. There are no common representations at the sub-word level.
Doc2Vec	A numeric representation of a document, regardless of its length. Faster than Word2vec.	The benefit of using doc2vec is diminished for shorter documents.
GloVe	GloVe, unlike Word2vec, does not rely solely on local statistics (Words local context information).	In order to obtain word vectors, global statistics (word co-occurrence) are used.
BERT	Identify and capture contextual meaning in a sentence or text.	Compute-intensive at inference time.

Table 1: Comparison of Different Text Representation Methods [5].

3.2.2 Word Embeddings

Unlike traditional vectorization methods, word embeddings provide dense vector representations of words in a continuous vector space [5]. Techniques like Word2Vec and GloVe (Global Vectors for Word Representation) capture semantic relationships between words, where similar words are closer in the vector space [5]. This is particularly useful for capturing the nuanced meanings in text, which can help in detecting subtleties in fake news.

These techniques basically provide dense vector representation of words in continuous vector space [5]. Word2Vec and GloVe (Global Vectors for Word Representation) capture the relationship between words in a vector space, where similar words are positioned closer [5]. This is beneficial for understanding the deeper meanings in text, which can assist in spotting the slight differences of fake news [5].

- **Word2Vec:** It basically uses neural networks to map words into a vector space based on their contexts in large corpora [5].
- **GloVe:** It uses how often words are placed together to create word representations, which makes it easier and faster to analyse how words are related to each other [5].

With the help of these NLP Techniques, researchers can develop more efficient and accurate models for detecting fake news.

4 Available Datasets for Fake News Detection

The list of available datasets for fake news detection is big, but a few most authentic resources are listed below:-

- **LIAR:** Contains manually labeled short statements from POLITIFACT.COM, known for its detailed analysis reports and links to source documents [6].

- **Fake News Detection Dataset (Kaggle):** Comprises articles from Reuter.com (true) and various fake news outlets, focusing on the period between 2016 and 2017 [6].
- **BuzzFeed News Dataset:** A collection of fake and real news articles from BuzzFeed News [6].
- **PolitiFact Dataset:** Contains fact-checked claims from PolitiFact [6].
- **WELFake:** A large and generic dataset created by merging four popular datasets to prevent overfitting [6].
- **MM-COVID:** Focuses on multilingual fake news related to COVID-19, providing both fake and trustworthy information in multiple languages [6].
- **Weibo21:** A benchmark for multi-domain fake news detection with domain labels [6].
- **Gossipcop Dataset:** Contains fact-checked claims from Gossipcop [6].
- **ISOT Fake News Dataset:** A dataset focused on fake news related to information security and online trust [6].

5 Comparison of Traditional vs LLM-based Methods

When it comes to traditional methods in detecting fake news, then there are numerous methods which are there. The traditional methods can be classified into manual and automatic approaches. The manual detection methods which were used earlier before the creation of computational methods include knowledge based methods, which are heavily dependent on fact - checking to verify information against the sources [7]. The other approach include style based approaches, where the text and the visualised content is analysed. this helps to identify potential differences with respect to credible sources. Propagation-based methods are also quite useful as they help in figuring out how news spread through social networks like Facebook, X(formerly known as twitter) and it also provides information regarding the unusual ways how the fake news is spread through social media platforms [7]. Another tradition method to check whether a news is fake or not , is the source-based methods, which investigates the credibility of the news origin and through which channels it was propelled into the public domain [7].

It also considers numerous factors while checking a news like author reputation and platform as well. Even though these traditional methods provide valuable information and insight , but still, the effectiveness of these methods is limited by numerous factors such as the evolving fake news tactics, data availability and how complex online information systems are [7]. The rapid growth of fake news on digital platforms poses a significant threat to society as it may lead to undermining trust in institutions and it influence public opinion in a negative way. With such rapid evolution of misinformation tactics, it is very difficult to curb to spread of fake news by traditional methods discussed above [7]. In order to detect fake news more effectively, the power of Large learning models(LLMs) can be put to use. LLMs are the very deep learning modules which are trained on vast amount of data. The model proposed in the study is DAFND, which addresses the challenges posed by fake news , by using the Large language models(LLMs), while incorporating a dual perspective [8]. DAFND is highly effective in detecting inconsistencies and can make a clear distinction between fabricated and credible information [8]. The results produced are highly effective and this tool basically has the potential to become an effective tool in inhibiting the spread of fake news and can also help to promote information literacy.

5.1 Survey of LLM-based Solutions for Fake News Detection

Even though there are numerous LLM methods which are there to tackle fake news . Two of such methods are discussed here in detail:-

- **DAFND Method**
- **SheepDog Method**

5.1.1 Model Architecture of DAFND and SheepDog Methods

DAFND Method: DAFND is highly effective when it comes to combat fake news. In order to find out, how does this help to detect fake news, we need to give a thorough outlook to it's model architecture. DAFND is composed of four interconnected modules:

- **Detection Module:** This is the initial step which involves the extraction of key information from the news article. The model basically captures the core content of the news piece by identifying the crucial keywords with the help of LLM. The extracted keywords are used as foundation for further analysis [8].
- **Investigation Module:** This module is extremely important in gathering information which relevant , from both external and internal resources [8].

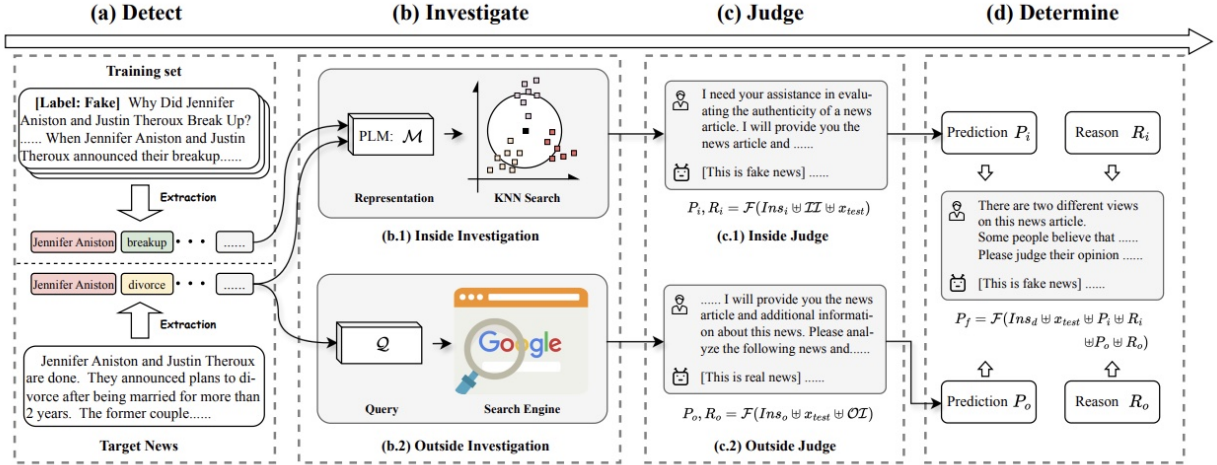


Figure 1: The architecture of our DAFND model. It includes four sequentially connected parts: (a) Detection Module (b) Investigation Module (c) Judge Module (d) Determination Module. [8]

- **Inside Investigation:** In order to address the challenge of uncertainty, the models gathers similar news articles from the training dataset. This process, is further incorporated by a similarity metric, which helps the LLM to understand the context of the target news in a better [8].
- **Outside Investigation:** In order to combat information scarcity, a real time search engine is being employed by the model to fetch real time information related to the news. This helps to verify the accuracy of the news and provides a broader perspective as well [8].
- **Judge Module:** This module basically evaluates information which was gather in the previous stage. LLMs are used to produce explanations and predictions , which is based on bot internal and external investigations. This is the step, where model attempts to determine the authenticity of the news from both perspectives [8].
- **Determination Module:** This is the the final component of this model, which combines the predictions and explanations from the previous module, which was judge module. A confident decision is made only , when both the perspectives align. However, if there is any contrast detected, then the model employs decision-making mechanism to evaluate the final classification [8].

As a whole DAFND’s strengths lies in effective utilization of LLM’s and a head on confrontation towards key challenges, which are currently there to tackle fake news. Even though it is robust and provide reliable results, but its performance is proportional to the LLM quality and Relies heavily on the quantity of training data.

SheepDog Method: The SheepDog Method is a style-agnostic fake news detection framework which is designed to reduce the impact of Large Language Model (LLM)-empowered style attacks [9]. These attacks basically involves changing the styles of fake news in order to make it look more authentic, which makes it very difficult for traditional fake news detectors to verify the authenticity and truthfulness of news articles [9]. The SheepDog Method addresses this challenge by focusing on content rather than style and use a combination of content focused truth evaluations and style-independent training methods [9]. Here’s a detailed explanation of the methodology and steps involved in the SheepDog method:

Step 1. LLM-Empowered News Reframing

To prepare for training, SheepDog first enhances the dataset with diverse stylistic variations:

- **Objective:** The objective is to generate training data that reflects a wide variety of stlistic variations and expressions, tus making the making the model more robuurst [9].
- **Process:** Each and every news article in the training set is passed through a process called **reframing** [9], where LLM are used to generate numerous varieties of stylistic versions of the article:
 - * Real news articles are rewritten in an unreliable and unauthentic style (e.g., “sensational”) [9].
 - * Fake news articles are rewritten in a reliable and authentic style (e.g., “objective and professional”) [9].
- **Output:** For every news article , SheepDog method creates two versions:

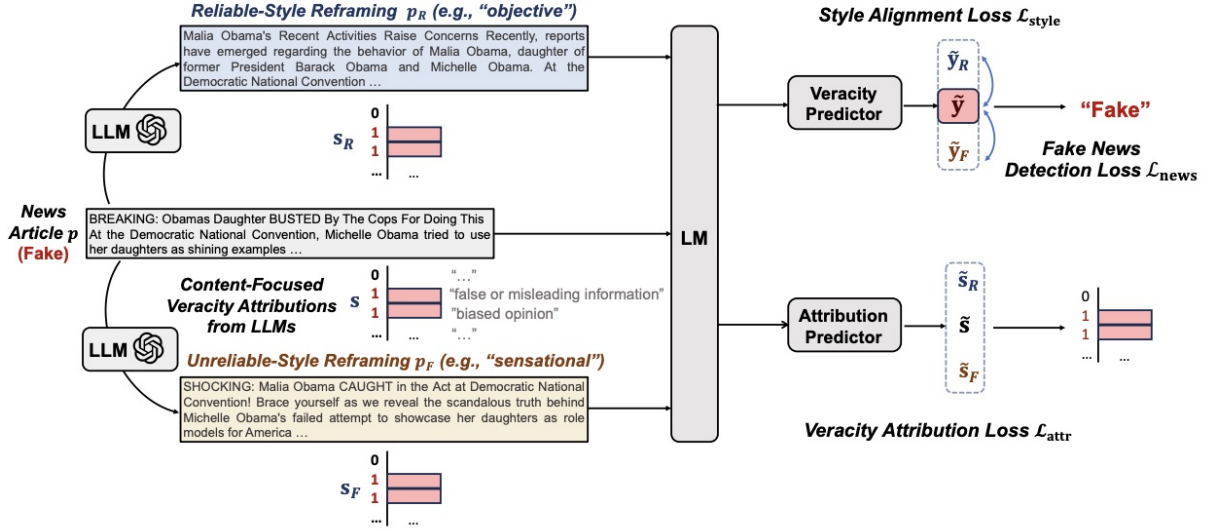


Figure 2: Overview of our SheepDog style-agnostic fake news detector [9].

- * **Reliable-style reframing** (p_R): It presents the article in a style which resembles more to reliable and authentic news source [9].
- * **Unreliable-style reframing** (p_F): It presents the article in a style which appears to resemble more towards the unreliable and unauthentic sources [9].

Step 2. Style-Agnostic Training

The main feature of this model is that it tries to focus on content rather than style by comparing predictions across different version of the same news article styled in a different way [9].

- **Style Alignment Objective:** The key idea here is to make sure that that prediction of the model remain constant regardless of the variation in style of the article [9].
- **Input Representations:** Here the models generates representations for the original news article (p), along with its reliable-style reframing (p_R), and the unreliable-style reframing (p_F) [9].
- **Loss Calculations:** In order to predict the contrast in predictions between the original and reframed versions, a style alignment loss ($\mathcal{L}_{style}$) is calculated. The model promotes style-agnostic veracity assessment, by minimising the divergence through training [9].

Step 3. Content-Focused Veracity Attributions

Another feature of this model is that it add additional content related information to improve the predictions:

- **Attribution Extraction:** LLMs are basically used to provide content focused truth checks, which involves:
 - * **Prompts:** The models basically asks LLM to identify characteristics or words , which are typically related to fake news, such as "lack of credible sources" or "misleading information [9]."
 - * **Pseudo-Labels:** The outputs which is provided by these prompts helps to figure out the presence or absence of certain fake news in the article [9].
- **Attribution Prediction Objective:** The model grounds its predictions solely based on content inspite of style, so the model is also trained to predict the psuedo -labels as well [9].

Step 4. Training Objectives and Loss Functions

In order to achieve the objectives, multiple loss functions are involved throughout the overall training process:

- **Style Alignment Loss ($\mathcal{L}_{style}$):** This loss function basically minimizes the Kullback-Leibler (KL) divergence between the predictions of the original and reframed articles, in order to ensure constant predictions among different styles [9].

- **News Detection Loss (L_{news}):** This loss function is a cross entropy loss function, which make sure, that the model predict the authenticity of the original news article in an accurate way [9].
- **Veracity Attribution Loss (L_{attr}):** this is a binary cross entropy loss function, which trains the model to predict the content-focused truth labels correctly [9].

Step 5. Final Objective Function

At last, the final objective function which is used to train the SheepDog model is a weighed combination of above mentioned losses:

$$L = L_{style} + L_{news} + L_{attr}$$

This loss function allows SheepDog to learn both content focused veracity indicators along with style agnostic representations, which make sure that even under LLM-empowered style attacks, there is reliable and robust fake news detection [9].

5.2 LLM-Empowered Style Attacks

LLM-Empowered Style Attacks includes the usage of large language models (LLMs) to modify or change the writing styles of news articles or information, which ultimately disguises fake news as a genuine source. This technique makes use of powerful generative and rephrasing capabilities of LLMs , such as GPT-3.5 or Llama2, which makes fake news appear more authentic and credible or to make real news look fabricated and untrustworthy, simply by adopting the linguistic characteristics of famous publishers [10]. For instance, a fake news article can be written to copy the style of reputable sources like "The New York Times", which makes it very difficult for readers and other automated detectors to verify it's authenticity. In addition to this, real news can be restyled in sensational manner, which may potentially lead to wrong classification as unauthentic content [10].

Moreover, the biggest challenge with catching fake news is that, traditional text-based fake news detectors usually look for clues how words are put together and try to find lies in it, but by altering these clues LLMs can significantly decrease the effectiveness of such detectors, which can be clearly seen by substantial decrease in F1 scores across various models, which faced such attacks [10].

6 Combating Fake News: Enhancing Societal Awareness and Governance through Education and Policy

6.1 Role of Education and Media Literacy

Media literacy plays a crucial role in fake news detection by making individuals skilled enough to evaluate the information in a critical manner [11]. The literature review done by Eisemann and Pimmer (2020) offers an overview of numerous media interventions which are designed in order to tackle and detect fake news [11]. Over two decades (2000-2020), Eisemann and Pimmerscreened almost 995 articles in order to identify the effective media literacy training and interventions [11]. After the screening, they found out a total of 19 articles were there, which met that specific criteria, which was comprised of solid methodology, evidence3-based methods, reported outcomes and inclusion of explicit fake news components [11]. They went on to identify 3 types of training:-

Reactive Training

This training basically targeted at correcting existing misconceptions through fact-based corrections. A total of nine articles were found which used this training method and involved strategies which were a mix of scientific evidence provision and dramatic narratives [11]. The overall effectiveness reduced with respect to time [11].

Proactive Training

This training basically involved the improvement of an individuals abilities to detect fake news using tools such as reverse image searches, cross checking and as fact checkers as well [11]. A Total of 11 articles had this method and was found that this method is more effective and sustained over time when it was compared with re4active training [11].

Critical Understanding of Media Systems

This method basically focuses on helping individuals and critically reflect the media content and its creation [11]. Even though, it was prevalent in scientific literature for more than 20 years, this approach showed limited effectiveness [11].

Overtime, news evaluation approaches has proven that there is a necessity to use tools by the individuals in order to clearly differentiate real news from disinformation. when it comes to media literacy, it can significantly increase the chances of individuals to combat fake news and it also lays stress on diversifying fake news detection techniques [11].

6.2 Implications for Public Policy and Governance

There have been numerous steps which have been taken by various media outlets and governments across the world to counter fake news [11]. In 2018, ahead of the European Parliament elections, the European Union (EU) tackled the disinformation at a massive scale in order to ensure more and more people engage in voting [11]. The documents analysed by Brites et al. include:

1. **Report of the Independent High Level Group on Fake News and Online Disinformation:** It basically proposes a set multidimensional approaches to combat fake news and disinformation [11].
2. **Commission Recommendation on Enhancing the 2019 European Parliament Elections:** This basically had sole focus on conducting the election within EU in an efficient way and to strengthen the European nature [11].
3. **EU Code of Practice on Disinformation:** This basically contained a set of guidelines for all advertisers and online platforms to counter fake news in a more efficient way [11].
4. **Tackling Online Disinformation: A European Approach:** It basically tells about specific approaches to counter the fake news within the EU borders [11].
5. **Fake News and Disinformation Online (Flash Eurobarometer 464):** This basically consists of surveys and which are based on public perception and how do the public understands and reacts, when they counter fake news [11].

Brites et al. concluded that these documents generally failed to tackle fake news from a generational point of view, as it treated both adults and younger people as one single group and didn't recognize specific needs and opportunities [11].

German Perspective on Disinformation

Germany's approach to disinformation involves several ministries and authorities working in parallel:

1. **Federal Foreign Office (AA):** This office basically treats disinformation as tools by the foreign actors and states. It also maintains a close dialogue with international partners like the EU, G7, and NATO [12].
2. **Federal Ministry of the Interior (BMI):** This ministry basically dictates the national strategy against hybrid threats, including online propaganda and fake news. It also ensures that there is a quick and decisive action taken against disinformation in shortest time possible and it also plays a strategic role in threat assessment and response [12].
3. **Federal Office for the Protection of the Constitution (BfV):** This office basically ensures that the democratic order remains intact and monitors all threats including cyber-operations and propaganda peddled by foreign actors. In order to carry out the tasks, it uses all sources and intelligence services available [12].

Combating Disinformation

In order to combat fake news quickly and effectively, rapid identification and action are crucial. Germany's Federal Government have a coordinated approach, with the BMI takes care of the task force, which monitors the fake news and facilitates inter-ministerial dialogue [12]. The government proactively informs and warns the public, even when there is acute risk [12].

Additionally, social media platforms have a responsibility to counter fake news. The **Network Enforcement Act (NetzDG)** [13] mandates quick processing of user complaints, while the **Digital Services Act** [14] at the European level enforces stricter regulations. These measures make sure that the monetization of the fake news/ disinformation is completely stopped and mechanisms which report fake news are enhanced [14] [13]. Overall, the coordinated efforts at both the EU and national levels underscore the importance of tailored strategies and quick responses to mitigate the impact of disinformation on public opinion and democratic processes.

7 Challenges which exist with respect to fake news

Deepfake Detection

There have been several challenges when it comes to combat fake news. The emerging technologies have made it difficult to combat fake news in fraction of second. One of the major challenge which exist when it comes to fake news is dealing with deepfakes. A deepfake is an AI-generated video or image, which looks realistic. It is very difficult to authenticate such media because of their complex nature [15].

Influencer Account Hacking

This is again a significant challenge as if a person's account is hacked, who is having large following, then it leads to propagation of fake news to a larger section of society in a very short span of time. Before the misinformation can be corrected, a lot of damage may be financial or status could happen [15].

Real-Time Detection

The dynamic nature of news also affects the fake news detection process. A news which may appear fake at one time, may become truth later on. Thus, real-time detection systems become necessity for countering fake news [15].

Benchmark Datasets and Standard Features

Till now, the creation of benchmark datasets and evaluating standard feature sets for fake news detection is still an unresolved challenge [15].

Combining Multiple Features for Detection

In order to produce most effective results for fake news detection, fake news detection methods needs to combine content, context, and propagation features, but it is very challenging to make it possible without high computational complexity [15].

Ambiguity and Information Scarcity in LLMs

Large Language Models (LLMs) face a lot difficulty in understanding and evaluating unclear content. LLMs also have to deal with the outdated training corpus due to the timeliness of news [8].

Social Bots and Echo Chambers

Social media bots, which are followed in large numbers, promote mostly false impressions, which further leads to the formation of echo-chambers, which peddles fake news at a massive scale [16].

8 Conclusion

In conclusion, the enormous problem of fake news creates complex set of problems, which requires advance strategies for detection and prevention. The description of numerous machine learning and NLP techniques, highlights the advancements in technologies for identifying misinformation. Even though, traditional methods are effective, but they are more time consuming as compared to advance methods, and can't be put to use for fake news detection, in short interval of time. The emergence of LLM-based solutions, exemplified by DAFND and SheepDog, are evident that such methods are highly effective, when it comes to fake news detection. Even though, these methods are effective, but still a challenge is posed by the LLM-Styled Attacks, which makes fake news look more reliable and authentic. Even though advance technologies are required, the role of education in fostering media literacy, accompanied by forward-thinking public policies, can't be ignored in order to make a strong information ecosystem. In addition to this, challenges like deepfake content, influencer account hacking, and the dynamics of social bots and echo chambers are also discussed. As a whole, the paper focuses on Technological innovation, educational initiatives, and policy interventions to moderate the impact of fake news and protect the interests of public and ensuring confidence in stabilised democratic values.

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